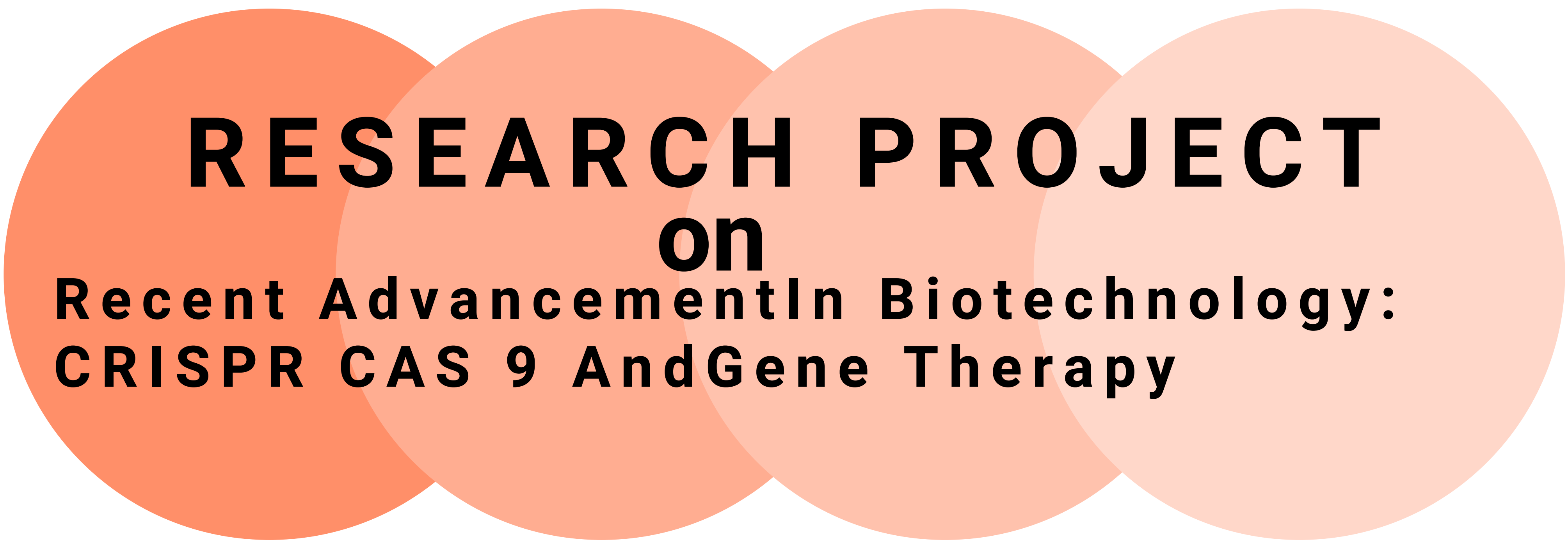




RESEARCH PROJECT

on

**Recent Advancement In Biotechnology:
CRISPR CAS 9 And Gene Therapy**

Research Notes

Topic:

Recent Advancements in Biotechnology: CRISPR-Cas9 and Gene Therapy

1. Introduction to Biotechnology

Notes:

Biotechnology is the application of biological systems and organisms to develop useful products and technologies.

Used in healthcare, agriculture, food production and environmental management.

Modern biotechnology includes genetic engineering and genome editing.

2. What is CRISPR-Cas9?

Notes:

CRISPR stands for Clustered Regularly Interspaced Short Palindromic Repeats

Cas9 is an enzyme that acts like molecular scissors.

Originally discovered as a bacterial defense system against viruses.

Scientists adapted CRISPR-Cas9 for precise genome editing.

3. Scientists Associated with CRISPR

Notes:

Jennifer Doudna

Emmanuelle Charpentier

Both Female Scientists Awarded with Nobel Prize in Chemistry (2020)

4. How CRISPR-Cas9 Works

Notes:

Guide RNA identifies the target DNA sequence.

Cas9 enzyme cuts the DNA.

Cell repairs DNA.

Scientists can remove, insert or modify genes.

5. Applications of CRISPR

- Medical Application
- Cancer treatment research
- Agricultural Applications
- Improved crop yield
- Industrial Application
- Enzyme engineering
- Gene therapy
- Rare genetic disorders
- Disease-resistant crops
- Drought-resistant plants
- Biofuel production

6. Gene Therapy

Notes:

- Treatment involving modification of genes.
- Can replace defective genes.
- Used for inherited disorders.
- CRISPR is making gene therapy more accurate.

7. Recent Advancements (2023-2026)

Notes:

- Progress in treatment of sickle cell disease.
- Improved precision gene editing.
- Development of next-generation CRISPR systems.
- Clinical trials for genetic disorders.

8. Advantages

Notes:

- High precision
- Faster than older techniques
- Cost-effective
- Broad applications

9. Challenges and Ethical Concerns

Notes:

- Off-target effects
- Ethical concerns regarding human embryo editing
- Regulatory issues
- Long-term safety concerns

10. Future Prospects

Notes:

- Personalized medicine
- Advanced gene therapies
- Sustainable agriculture
- Disease prevention

11. References to Collect

Sources Needed:

- NIH
- WHO
- Nature Reviews Genetics
- ScienceDirect
- Nobel Prize Website